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Brightlearn

Exercise no 1

SQL Fundamentals

01 Selecting and Sorting

1. All columns

```
24 | SELECT *
25 | FROM spar.profiles.employees;
26 |
27 |
28 |
29 |
30 |
31 |
```

Generate (Ctrl + I)

Add parameter

Table +

	id	first_name	last_name	department	salary	hire_date	city
1	1	Alice	Green	IT	70000	2020-01-10	Johannesbu...
2	2	Brain	Lee	HR	45000	2019-03-22	Cape Town
3	3	Cathy	Zulu	Finance	65000	2018-07-18	Durban

5 rows 0.637s runtime [See performance \(1\)](#) ✓ Checked for improvements Refreshed Yesterday

2. Unique department

```
27 | --- distinct departments
28 | SELECT DISTINCT department
29 | FROM spar.profiles.employees;
```

Add parameter

Table +

	department
1	IT
2	HR
3	Finance
4	Marketing

3. First and last name ordered by salary descending

```
30 | --- first and last name ordered by salary descending
31 | SELECT first_name,
32 |        last_name
33 | FROM spar.profiles.employees
34 | ORDER BY salary DESC;
```

Add parameter

Table +

	first_name	last_name
1	Eva	Naidoo
2	Alice	Green
3	Cathy	Zulu
4	David	Mokoena
5	Brain	Lee

4. Top 3 highest paid employees

```
35 --- top 3 paid employees
36 SELECT id,
37        first_name,
38        last_name,
39        salary
40 FROM spar.profiles.employees
41 ORDER BY salary DESC
42 LIMIT 3;
43
```

add parameter

	id	first_name	last_name	salary
1	5	Eva	Naidoo	72000
2	1	Alice	Green	70000
3	3	Cathy	Zulu	65000

02 Filtering with WHERE

5. Employees in IT department

```
44 --- employees in IT department
45 SELECT id,
46        first_name,
47        last_name,
48        department
49 FROM spar.profiles.employees
50 WHERE department = 'IT';
51
```

add parameter

	id	first_name	last_name	department
1	1	Alice	Green	IT
2	5	Eva	Naidoo	IT

6. Employees in finance and salary > 60000

```
51 --- employees in finance with salary > 60000
52 SELECT id,
53        first_name,
54        last_name,
55        department
56 FROM spar.profiles.employees
57 WHERE department = 'Finance' AND salary > 60000;
58
```

add parameter

	id	first_name	last_name	department
1	3	Cathy	Zulu	Finance

7. Employees in HR or Marketing

```

8 --- employees in HR or marketing
9 SELECT id,
10     first_name,
11     last_name,
12     department
13 FROM spar.profiles.employees
14 WHERE department= 'HR' OR department= 'Marketing';
15

```

Add parameter

id	first_name	last_name	department
2	Brain	Lee	HR
4	David	Mokoena	Marketing

8. Employees not in IT

```

65 --- employees NOT in IT
66 SELECT id,
67     first_name,
68     last_name,
69     department
70 FROM spar.profiles.employees
71 WHERE department != 'IT';

```

Add parameter

id	first_name	last_name	department
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

9. Employees IN IT, HR OR Finance

```

72 --- employees in IT, HR or Finance
73 SELECT id,
74     first_name,
75     last_name,
76     department
77 FROM spar.profiles.employees
78 WHERE department IN ('IT', 'HR', 'Finance');
79
80

```

Add parameter

id	first_name	last_name	department
1	Alice	Green	IT
2	Brain	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10. Employees in IT with salary > 65000 and city Johannesburg

```

79  --- employees in IT, salary > 65000 and city = 'Johannesburg'
80  SELECT id,
81         first_name,
82         last_name,
83         department
84  FROM spar.profiles.employees
85  WHERE department = 'IT' AND salary > 65000 AND city = 'Johannesburg';
86

```

Add parameter

Table +						🔍 🔧 📄 📊 📌 ⬆️ ✖️	
	1_3^2 id	Δ_C^B first_name	Δ_C^B last_name	Δ_C^B department			
1	1	Alice	Green	IT			
2	5	Eva	Naidoo	IT			